

GUAM INTL, Guam

WMO: 912120

Lat: 13.483N

Lon: 144.800E

Elev: 77

StdP: 100.40

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 41415

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.3	23.8	18.6	13.5	26.5	19.3	14.2	26.6	11.4	28.2	10.6	28.2	3.4	50	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	5.0	31.9	25.9	31.5	25.8	31.2	25.7	27.6	29.7	27.2	29.4	27.0	29.3	6.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.9	28.5	26.6	22.4	28.4	26.2	21.8	28.3	88.7	29.9	87.1	29.4	85.6	29.3	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.1	8.3	21.6	33.1	0.8	0.9	21.1	33.7	20.6	34.2	20.1	34.7	19.6	35.3		
(5)			19.4	28.0	0.9	1.0	18.8	28.8	18.2	29.4	17.7	30.0	17.0	30.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	27.0	26.9	27.4	28.0	28.4	28.5	28.0	27.6	27.6	27.8	28.1	27.7	(6)
(7)		DBStd	0.99	0.79	0.76	0.79	0.73	0.79	0.94	1.01	0.97	0.97	0.96	0.86	0.80	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6480	527	474	538	541	572	555	557	546	527	552	543	549	(10)
(11)		CDD18.3	3438	268	241	279	291	313	305	299	288	277	293	293	291	(11)
(12)		CDH23.3	36398	2481	2183	2744	3168	3602	3601	3320	3031	2871	3140	3220	3037	(12)
(13)		CDH26.7	10596	540	477	704	972	1231	1301	1049	849	789	913	978	793	(13)
(14)	Wind	WSAvg	4.3	5.2	5.4	5.3	5.1	4.3	4.0	3.3	3.1	2.9	3.4	4.8	5.3	(14)
(15)	Precipitation	PrecAvg	2288	110	91	72	89	145	159	276	360	341	301	210	136	(15)
(16)		PrecMax	3346	460	352	236	388	611	339	757	948	618	641	516	405	(16)
(17)		PrecMin	1471	24	8	10	13	10	21	132	99	188	109	70	40	(17)
(18)		PrecStd	481	91	77	52	81	129	79	119	168	116	109	91	74	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	30.7	31.2	31.7	32.1	32.2	32.2	31.9	31.9	31.9	31.7	31.2	(19)
(20)			MCWB	24.6	24.3	24.2	25.0	25.5	25.8	26.2	26.6	26.5	26.4	26.0	25.7	(20)
(21)		2%	DB	30.2	30.2	30.7	31.2	31.6	31.7	31.4	31.1	31.1	31.2	31.2	30.7	(21)
(22)			MCWB	24.5	24.3	24.3	24.9	25.5	25.8	26.1	26.3	26.3	26.2	25.8	25.4	(22)
(23)		5%	DB	29.8	29.8	30.2	30.8	31.1	31.2	30.9	30.6	30.6	30.7	30.8	30.3	(23)
(24)			MCWB	24.4	24.2	24.2	24.9	25.5	25.8	26.0	26.2	26.2	26.1	25.8	25.3	(24)
(25)		10%	DB	29.4	29.3	29.8	30.4	30.7	30.8	30.4	30.1	30.0	30.2	30.3	29.8	(25)
(26)			MCWB	24.3	24.1	24.2	24.9	25.4	25.8	26.0	26.2	26.2	26.1	25.8	25.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.6	26.6	27.1	27.5	27.4	27.8	27.8	27.7	28.1	27.9	27.7	(27)
(28)			MCDWB	28.3	28.2	28.5	29.3	29.8	29.8	29.9	30.1	29.8	29.9	29.4	29.4	(28)
(29)		2%	WB	26.0	25.9	25.8	26.4	26.9	27.0	27.2	27.2	27.2	27.4	27.2	27.0	(29)
(30)			MCDWB	28.2	27.9	28.2	29.1	29.4	29.7	29.4	29.5	29.5	29.5	29.3	28.8	(30)
(31)		5%	WB	25.6	25.4	25.4	26.0	26.5	26.6	26.8	26.8	26.8	27.0	26.8	26.4	(31)
(32)			MCDWB	27.9	27.7	28.1	29.0	29.3	29.5	29.2	29.2	29.2	29.3	29.1	28.6	(32)
(33)		10%	WB	25.2	25.0	25.0	25.7	26.1	26.2	26.4	26.4	26.4	26.5	26.3	25.9	(33)
(34)			MCDWB	27.6	27.6	27.9	28.7	29.2	29.4	29.1	28.9	28.9	29.0	28.9	28.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.2	5.4	5.3	5.1	5.0	5.1	5.2	5.2	5.1	4.9	4.8	(35)
(36)			MCDBR	5.6	5.8	5.8	5.6	5.4	5.4	5.6	5.8	5.9	5.7	5.3	5.2	(36)
(37)			MCWBR	2.3	2.3	2.2	2.0	1.9	2.0	2.3	2.6	2.5	2.5	2.3	2.2	(37)
(38)		5% WB	MCDWB	4.8	5.1	5.1	5.1	5.0	5.0	5.2	5.4	5.4	5.2	4.9	4.7	(38)
(39)			MCWBR	2.5	2.6	2.4	2.2	2.2	2.2	2.5	2.7	2.7	2.7	2.5	2.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.412	0.413	0.412	0.412	0.412	0.409	0.404	0.408	0.395	0.421	0.437	0.427	(40)	
(41)		taud	2.471	2.469	2.485	2.512	2.536	2.547	2.576	2.555	2.603	2.511	2.416	2.422	(41)	
(42)		Ebn at Noon	887	901	906	897	882	875	882	888	908	882	856	861	(42)	
(43)		Edh at Noon	109	113	113	110	105	103	100	104	99	107	114	112	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.96	5.64	6.36	6.82	6.67	6.38	5.62	5.16	5.12	5.05	5.11	4.90	(44)	
(45)		RadStd	0.27	0.23	0.27	0.26	0.27	0.50	0.54	0.46	0.39	0.47	0.27	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.26	+0.21	N/A	+0.19	N/A	N/A	N/A	N/A	+83	+82
(47) Regional (1 neighbor)		+0.16	+0.14	N/A	+0.15	N/A	N/A	N/A	N/A	+61	+62

Nomenclature: See separate page